PROJECT CASE STUDY

SAFISH PLUS POS

A desktop fish-retail system connecting weight-based sales, FIFO stock batches, supplier restocks, and operational reporting in one English–Urdu workspace.

SYSTEM	SCOPE	STATUS
Fish Retail & Point of Sale	**Sales · Stock · Suppliers Operational Reporting**	**Implemented Desktop Application**

01 The Problem

Fish retailers sell by weight while purchase costs vary between deliveries. Each checkout needs a chosen selling rate, an available-stock check, and a clear invoice. Behind the counter, the same business must follow which batches remain, which suppliers delivered them, and what was earned from daily sales.

Keeping sales, restock notes, stock balances, and supplier records separately makes it harder to inspect remaining inventory and review trading performance from the same history.

> The application connects checkout, supplier-linked stock batches, invoices, and reports in a locally stored desktop workflow.

02 What Was Built

A bilingual point-of-sale and inventory application for fish retail, with persistent business records and a dedicated desktop interface.

WEIGHT-BASED CHECKOUT

Build a multi-item cart in kilograms, enter selling prices, apply an invoice discount, and record cash or online payment classification.

INVENTORY & FIFO BATCHES

Maintain fish records, receive stock from suppliers, and track remaining quantities and purchase costs for individual deliveries.

SUPPLIER RESTOCK HISTORY

Store supplier contact and shop details. Review delivery counts, cumulative restock spend, and the most recent receipt of stock.

REPORTING & RECEIPTS

Review six report views, save five report types as CSV, and generate invoice receipt PDFs for viewing and printing.

03 How It Works

Users enter and review transactions in the desktop workspace. Named requests cross a context-isolated preload bridge to main-process handlers, which validate inputs and read or update a local SQLite database.

React workspace → Preload bridge → IPC handlers → SQLite

Supplier restock → Inventory quantity + dated cost batch
Checkout → FIFO deductions → Invoice + sale lines

Receive fish & build stock

Create a fish record, then select a supplier and enter the delivered weight and purchase rate. Restocking updates the item quantity and average purchase price while preserving a dated batch with its own remaining quantity and cost.

Find products & prepare the cart

Search fish by name or numeric ID. Sales search combines fuzzy matching with English/Roman–Urdu alternatives. Add available products, adjust kilogram quantities and selling rates, and review the subtotal and invoice discount.

Post the sale & produce a receipt

Checkout checks availability and commits batch deductions, inventory changes, the invoice, and its sale lines in a database transaction. Costing consumes older active batches first; invoice totals and profit account for the discount.

Review operations & share output

The dashboard surfaces daily revenue, transaction count, stock value, alerts, and activity. Reports query stored history; selected views export to CSV. Invoice actions generate a receipt PDF and open it in the system viewer for printing.

04 Key Features

- **Weight-based multi-item sales** — kilogram quantities and manually entered selling rates feed the cart. Checkout stores numbered invoices and linked fish-sale lines.

- **Supplier-linked restocking** — deliveries retain supplier, fish, date, received weight, remaining weight, purchase rate, and total purchase amount.

- **FIFO-based sale costing** — checkout consumes older available batches and uses their costs to calculate line profit. Stock and sale writes are committed together.

- **Discounts & payment classification** — invoice discounts reduce the final amount and invoice profit. Cash/online labels record the selected category, not a live payment integration.

- **Inventory visibility** — fish lists, quantity and value summaries, configurable warning/critical thresholds, and low-stock dashboard alerts support stock review.

- **Six operational report views** — daily sales, sales history, inventory, product performance, trends, and supplier restocks provide date-based or rolling-period analysis where applicable.

- **CSV reports & PDF receipts** — five report types save structured CSV data. Bilingual invoice receipts show items, totals, discounts, and configured shop details in a PDF.

- **English–Urdu desktop workflow** — localized labels, right-to-left layout, cross-script sales search, and keyboard shortcuts support counter use. Stored names remain as entered.

05 Technical Approach

The implementation separates renderer screens, the preload API, main-process business requests, and local persistence. Electron supplies the desktop runtime; Vite builds the React interface.

LAYER	STACK	PURPOSE
Desktop runtime	Electron 40 · Node.js	Window lifecycle, named IPC requests, native dialogs, and receipt PDF generation.
Frontend & tooling	React 19 · JavaScript · Vite 7	Component-based screens, local UI state, lazy-loaded views, and renderer bundling.
Interface & analytics	Mantine 8 · Recharts · Fuse.js	Forms, notifications, report charts, and fuzzy fish search at checkout.
Data persistence	SQLite · better-sqlite3	Local records, prepared queries, WAL-mode operation, and transactional sale posting.
Localization	i18next · react-i18next	English/Urdu catalogs, cached language choice, RTL presentation, and translated labels.
Output & packaging	Electron printToPDF · CSV worker · electron-builder	Receipt PDFs, UTF-8 CSV reports, and Windows NSIS installer configuration.

Design decisions that matter

A narrow renderer-to-main boundary. The UI calls domain-specific functions exposed through the preload bridge instead of issuing renderer-side SQL. Database access and native output stay in the main process.

Batch history alongside invoice records. Restock batches retain delivery costs and remaining quantities; sale lines link fish to invoices. The checkout transaction updates the records together.

Explicit desktop scope. This is a local retail POS, not a customer-ledger, consignment-settlement, cloud-sync, or payment-processing service. CSV report exports and receipt PDFs are separate output paths.

06 Value Delivered

From separate counter records to a connected, inspectable fish-retail workflow:

Connected	Traceable	Bilingual
SALES & STOCK	BATCHES & INVOICES	LOCAL OPERATIONS

- ✓ **One operational workspace.** — Checkout, fish records, supplier deliveries, stock review, receipts, and reports share an application and a local database rather than disconnected screen-only data.
- ✓ **Inspectable stock and costing.** — Supplier-linked batches expose what was received and what remains, while FIFO sale costing connects the checkout calculation to available delivery costs.
- ✓ **A clearer day-end view.** — Daily invoice totals, transaction counts, discounts, payment labels, product performance, and restock reports make trading activity available for review.
- ✓ **A practical local-language interface.** — English and Urdu labels, right-to-left presentation, keyboard-assisted checkout, and native file output suit a locally operated retail counter.

SAFISH PLUS POS — a local-first desktop point-of-sale system connecting weight-based checkout, batch-aware inventory, supplier restocks, and operational reporting for fish retailers.